

### Advanced (Habanero)

**Q1.** Factor the expression  $x^3 + 8$

- (A)  $(x + 2)(x^2 - 2x + 4)$
- (B)  $(x - 2)(x^2 + 2x + 4)$
- (C)  $(x + 2)(x^2 + 2x + 4)$
- (D)  $(x - 2)(x^2 - 2x - 4)$
- (E)  $(x + 2)(x^2 - 2x - 4)$

**Q2.** Factor the expression  $27x^3 - 8y^3$

- (A)  $(3x - 2y)(9x^2 + 6xy + 4y^2)$
- (B)  $(3x + 2y)(9x^2 - 6xy + 4y^2)$
- (C)  $(3x - 2y)(9x^2 - 6xy + 4y^2)$
- (D)  $(3x + 2y)(9x^2 + 6xy + 4y^2)$
- (E)  $(3x - 2y)(9x^2 + 6xy - 4y^2)$

**Q3.** Recognize the cube pattern in  $x^3 - 64$

- (A)  $x^3 - 4^3$
- (B)  $x^3 + 4^3$
- (C)  $x^3 - 8^3$
- (D)  $x^3 + 8^3$
- (E)  $x^3 - 3^3$

**Q4.** Factor the expression  $8x^3 + 27y^3$

- (A)  $(2x + 3y)(4x^2 - 6xy + 9y^2)$
- (B)  $(2x - 3y)(4x^2 + 6xy + 9y^2)$
- (C)  $(2x + 3y)(4x^2 + 6xy + 9y^2)$
- (D)  $(2x - 3y)(4x^2 - 6xy + 9y^2)$
- (E)  $(2x + 3y)(4x^2 - 6xy - 9y^2)$

**Q5.** Simplify  $(x + 2)(x^2 - 2x + 4)$

- (A)  $x^3 + 8$
- (B)  $x^3 - 8$
- (C)  $x^3 + 2x - 4$
- (D)  $x^3 - 2x + 4$
- (E)  $x^3 - 2x - 4$

**Q6.** Factor  $x^3 - 125$

- (A)  $(x - 5)(x^2 + 5x + 25)$
- (B)  $(x + 5)(x^2 - 5x + 25)$
- (C)  $(x - 5)(x^2 - 5x + 25)$
- (D)  $(x + 5)(x^2 + 5x + 25)$
- (E)  $(x + 5)(x^2 - 5x - 25)$

**Q7.** Recognize the cube pattern in  $64x^3 + 27y^3$

- (A)  $4^3x^3 + 3^3y^3$
- (B)  $4^3x^3 - 3^3y^3$
- (C)  $8^3x^3 + 3^3y^3$
- (D)  $8^3x^3 - 3^3y^3$
- (E)  $2^3x^3 + 3^3y^3$

**Q8.** Simplify  $(x - 2)(x^2 + 2x + 4)$

- (A)  $x^3 + 8$
- (B)  $x^3 - 8$
- (C)  $x^3 + 2x - 4$
- (D)  $x^3 - 2x + 4$
- (E)  $x^3 - 2x - 4$

#### Open Ended Questions (FRQ)

**Q9.** Factor the expression  $x^3 - 64y^3$  and show your work step-by-step

**Q10.** Factor the expression  $27x^3 + 8y^3$  and simplify your answer

#### Chef's Tips

- Provide examples of factoring sum and difference of cubes
- Encourage students to simplify expressions after factoring
- Remind students to check their work by multiplying the factors